A Speech and a Train

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The Urda Disaster of 28 October 2026 — and Everything Around It

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An independent research document, written from the public record.

Every fact in this document comes from a named source — court documents, official bulletins, government press releases, regulator rulings, wire services, and newspaper reporting — and the source is noted as we go. Where sources disagree, both versions are written down and you get to see the disagreement. Where a claim turned out to be false, it is still written down, because a false claim circulating is itself a fact worth documenting.

Nothing in this document decides who is guilty. That is the job of the judge in Orgaz, the technical investigators, and the courts. This document only lays out everything that is known, in full, in plain language, so that anyone — including someone who has never read a word about trains — can understand the whole story from beginning to end.

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How to Read This Document

Imagine a friend who knows nothing about trains, Spain, or railway signals asks: "So what actually happened at Urda, and why does everyone say it was impossible?" This document is the answer you would give them over a long evening — with every detail, explained as you go.

A few ground rules the document follows:

We name names, but we do not blame. People are named because they held positions, made statements, signed contracts, or appeared in records. Naming them is not accusing them. The operator in the control centre, the minister, the company presidents, the judge — all are presented with exactly what the record says about them, no more and no less.

Every number has a history. You will see the death toll change hour by hour. You will see passenger counts that differ by twenty depending on who counted. Rather than silently picking one, this document shows the whole set, and tells you which one is official and why.

False claims are documented as claims. In the days after the crash, fake videos and fabricated quotes spread online — including old footage of the Israeli prime minister that was re-labelled with threats he never made. Those fakes are part of the story of what happened after the crash, so they are documented — as fakes.

The speech and the train are one story. This document is built on two threads: a speech given in April 2026, and a crash in October 2026. They are told in the order they entered the public

record — the crash first, at the front of the book, and the speech when it comes back into the story, which is exactly where, and when, it came back in real life: in the strange days after the crash, in a way that reveals a great deal about both. Both threads are told in full, and by the end they will be one.

With that, let us begin where the day began: before sunrise, in Madrid.

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PART ONE — THE DAY

1. Two ordinary trains on an ordinary morning

Tuesday, 28 October 2026, was shaping up to be a normal busy morning on Spain's oldest high-speed line.

At 07:54, a train belonging to the private operator Iryo pulled out of Madrid's Atocha station. Its number was 6076. It was a Frecciarossa 1000 — in Spain it carries the series number 109 — built by the Japanese-Italian partnership of Hitachi and Bombardier, with eight carriages and a sleek silver nose. On board were 147 passengers and five crew: three attendants, a barista, and the driver, a woman whose name has never been published and who, as you will see, plays one of the most important roles in this story. Her train was headed for Seville.

Six minutes later, at 08:00, a second train left the same station on the same track, bound for the same city. This was train 2080, run by the state operator Renfe — an AVE of the Series 103, a Siemens Velaro, the workhorse of Spanish high-speed rail since the mid-2000s. On board, according to the final official count by the accident investigators, were 379 passengers and seven crew. At the controls was a 36-year-old driver named Antonio Acedo.

A third operator's train — Ouigo service 6476, Madrid to Seville — had cleared the same corridor minutes earlier without incident, arriving punctually. It would be the last train to do so for weeks.

Both trains were on the Madrid–Seville high-speed line, the first of its kind in Spain, opened in April 1992 — the line on

which the very first AVE ran. It stretches 470.5 kilometres (about 292 miles) through the flat plains of Castilla-La Mancha, past Toledo and Ciudad Real, down through the Sierra Morena mountains to Córdoba, and on to Seville. On a normal morning, both trains would cover the distance in about two and a half hours at speeds of up to 300 kilometres per hour (186 mph), gliding under a protective umbrella of technology that — everyone believed — made a rear-end collision between two trains on the same track essentially impossible.

This was not a normal morning.

2. 08:30 — The system falls

At around 08:30, the computer system that supervised this entire stretch of line simply fell over.

The system was called LZB — the German acronym for a continuous train-control technology dating from the 1960s and 1970s. Think of it as an invisible electronic fence stretched along the whole railway: a cable between the rails talks constantly to each train, telling it how fast it may go, how far ahead the line is clear, and automatically slamming on the brakes if anything is wrong. Spain's 1992 line was built with it, and it had run for thirty-four years.

At the Urda signalling centre — the "central" that supervised the line around the small station of Urda, in the province of Toledo — three computers ran the LZB. When at least two of them stop agreeing, the system deliberately takes itself offline rather than risk giving wrong orders. That is what happened at 08:30, possibly a minute earlier or later. The computers

quarrelled, and the LZB across roughly sixty-three kilometres (about 39 miles) — some early reports said sixty, some sixty-seven, and we will meet these wobbling numbers again — went dark.

This was not, by itself, a catastrophe. The LZB was designed to fail safe. When it falls, trains are automatically stopped, and that is exactly what happened: six trains on the line braked to a halt, including both of ours. The driver of the Iryo, farther down the line, brought her train to a standstill before a red entry signal at Urda station. A few miles behind her, the AVE stopped at Mora. Both drivers did their jobs perfectly.

What happened next is the heart of this entire document. Between 08:30 and 08:54 — twenty-four minutes — a chain of human decisions, technical failures, and missed rescues unfolded, and none of the five safety barriers that should have stopped it did.

But first, the full picture of the technical mess, because the LZB failure was only the beginning.

3. The four failures

Within hours, journalists at the newspaper ABC obtained Adif's internal incident report — the "parte" that the duty staff file when something breaks. Dated 09:39 that same morning, it described not one failure but a cascade:

1. **The LZB central at Urda fell**, with what the report called "generalised transmission failures" in the areas known as Meta 40 and Meta 0. We will explain what those words mean later; for now, understand that the computers could no

longer tell trains even the most basic things — "this section is free" or "stop."

2. **The GSM-R radio — the railway's own mobile network** — began failing at points along the stretch. This was the channel drivers and controllers used to talk to each other.
3. **The track circuits at Urda station** — the devices that detect whether a stretch of rail is occupied by a train — began losing information intermittently.
4. **The electrical power at the exit of Urda station failed on both tracks**, and a track circuit called CV237 showed "occupied."

An Adif spokesperson would later insist that all communications with drivers "were made without problems" and that the failures had no direct bearing on the accident. The ministry, asked about the cascade, replied that "everything worked perfectly." Whether those reassurances survive the investigation is one of the open questions this document returns to at the end. What is certain is what the report itself says: four systems, one morning, one stretch of railway.

4. 08:31–08:36 — Confusion in the control room

In Madrid, in the Atocha control centre — the room with the big screens that Puente would later stand in front of, watching the whole disaster replay like a film — the night turned into improvisation.

The recordings, which would leak to three newspapers that same evening, preserve the atmosphere. You can hear the operators debating whether to even restart the system: "We're

deciding whether to disconnect or not." And then, presciently, one of them says: "And it's going to fail again."

At 08:36, the control centre telephoned the AVE driver and gave him an order, read from the rulebook's procedures for exactly this situation. In its full, banal form it went something like: "Driver of train 2080, disconnect and reconnect the LZB and resume your journey under BSL protection with normal running."

The driver repeated the critical part back, as drivers are trained to do: "With normal running?"

"Yes," said the controller.

That exchange — preserved on tape — would be replayed for days. Because "normal running" meant the driver could accelerate back up to line speed, around 200 kilometres per hour (about 124 mph) or more. And what the rulebook required instead was something called marcha a la vista — "running on sight" — a maximum of 40 kilometres per hour (25 mph), with the driver obliged to stop before anything he could see. The rulebook was unambiguous: with the line in this degraded state, with a red signal ahead, and with another train somewhere in the same section, "normal running" was not one of the options.

Experts consulted by the newspaper El País believe they know the precise technical mistake: the controller dictated the wrong category of telephone order. There are two nearly identical orders in the rulebook — LZB 3 and LZB 4 — differing essentially in whether the train must proceed at sight-seeing speed or may run normally. The LZB 4 was dictated; the situation called for LZB 3. The names are dissimilar enough to a

careful reader and similar enough to confuse a stressed one.

And the room was stressed. The recordings capture the operator garbling train numbers — the stationary Iryo becomes "6071-76," and then, in another fragment, "the 2061, that's in Urda" (a different AVE, coming the other way, which we will meet again). In the background someone admits the room's state of mind in one sentence: "Honestly, man, honestly… the 6071-76 is having problems with the equipment."

In that same window, a driver on another of the six halted trains did something remarkable: he pushed back. Told to resume with "normal running," he replied that it could not be normal running — it had to be marcha a la vista. The controller agreed, corrected the order for him ("Ah, yes, marcha a la vista"), and that driver proceeded safely. The correction was never passed on to the AVE. One driver saved himself with the rulebook; the rulebook did not save the next train.

5. 08:41 and 08:48 — The Iryo calls twice

The Iryo driver, meanwhile, was stranded. Her train refused to restart its LZB equipment — she tried, twice — and there she sat, before a red signal at the entrance to Urda station, a heavy silver train full of people standing on the main line.

At 08:41 she called Madrid. "The train won't let me restart the LZB," she reported, in the words preserved by Cadena SER, "and that's why I'm still stopped at the entry signal."

The control room's response, in full: "Okay, well, whenever you can, alright? And if you have problems, call us."

At 08:48 she called again. Still stopped. The equipment still would not restart.

Here is the fact that turned a railway accident into a national trauma: in the Atocha control centre, on the videographic screens, both trains were visible the whole time. The minister, Óscar Puente, would say he himself watched the "moviola" — the replay — and that "the trains are perfectly located at all times." The control room had been told twice, in real time, that a train was standing at Urda. And in the five to six minutes after the second call, as the AVE covered the last miles at speed, nobody warned the AVE. No counter-order exists in the recordings. The controller who had taken the call acknowledged it — and did nothing with it.

Why? Possibly because the room had talked itself into a wrong picture: the leaked chatter records the operators believing the Iryo was moving — "it's coming along in journey, it had a Meta 40 to continue in BSL" — while it had never moved at all. A red signal had been set to green in front of the stationary Iryo, because the room had prepared its exit route, expecting it to depart. So the picture in the cab of the oncoming AVE was, impossibly: a green signal ahead.

The AVE driver, Antonio Acedo, had himself reported a "partial block 00" — the most restrictive warning code, which means, in rulebook language, "the track to the next station is occupied." He knew, from his own instruments, that there was a train ahead. He obeyed the order anyway. Why does a driver do that? Because a driver cannot see the other train — the line curves, the terrain hides it — and because the whole

architecture of the railway is built on one assumption: that the person in the control room knows where every train is. "Surely," as one reconstruction put it, "the driver assumes that Atocha knows what it is doing."

6. 08:53 — The collision

At kilometre point 118 (mile 73), on the Madrid side of Urda station — technically in the municipality of Los Yébenes, which is why the official name of the accident is "the Los Yébenes accident," though the press mostly said "Urda" — the AVE caught up with the stationary Iryo.

The impact speed is estimated between 160 and 200 kilometres per hour (100 to 124 mph). The driver had seen the train ahead at the last moment and pulled the emergency brake; he died at his post, doing the only thing left to do. He was 36 years old, the son of a train driver and the grandson of a Renfe electrician, a photographer, a member of a small political party called Izquierda Española, and, by a twist this document will return to, the author of his party's transport programme, which had warned — in his own words — of "the glaring lack of maintenance funding and the abandonment of all these years translated into a demented deterioration of the rail infrastructure." His name was Antonio Acedo Meneses, and his self-written epitaph on his blog read: "Beauty is fleeting, fleeting on a train."

Physics did the rest. The eight-carriage AVE, nearly 200 metres of steel, buried its nose into the rearmost carriage of the Iryo. Both end carriages — the Iryo's tail car and the AVE's head

car — were smashed into each other, overturned, and spun down the embankment at the foot of the line. The rest of the AVE jack-knifed and derailed; its seventh carriage went down the slope, and a length of torn rail, ripped from the track, pierced one carriage like a javelin. Witnesses in the fields heard the impact miles away. A passenger, Juan Casares, thrown from the first carriage into the second, would describe breaking a window of the half-overturned car to climb out: "A lot of people aren't going home today, and I can't understand why."

The first photograph of the disaster was taken by a neighbour at 08:54.

7. The response — and the toll

Spain's emergency machinery, for all its later criticisms, worked that morning with everything it had. The regional emergency coordination centre declared the highest alert at 09:15. Around a hundred soldiers of the Emergency Military Unit, hundreds of Guardia Civil, dozens of fire crews, three medical helicopters and two intensive-care road units converged on a field in rural Toledo. The Guardia Civil's criminalistics teams flew drones over the wreckage to map it in three dimensions; dog teams searched the mangled cars through the night. The Guardia Civil's central operations unit — the force's elite investigative arm, deployed on the Adamuz model — reinforced the Toledo teams. Hospitals in Toledo, Ciudad Real, Madrid and beyond opened their doors; roughly two thousand people queued to give blood in the days that followed.

Within hours, sixty-four families had filed missing-person reports at Guardia Civil stations. The Guardia Civil's first working theory, recorded that afternoon, was that the Iryo had been stationary on the line for between five and ten minutes before the impact. And the mayor of Urda, Manuel Galán, who knew that stretch of line as well as anyone alive, kept returning to one fact in his interviews: it was a straight stretch, with good visibility — the kind of place where, under a clear morning sky, the questions about speed and signals and sightlines would all have to be answered honestly.

The small town of Urda, about 1,500 souls, turned its municipal pavilion into an impromptu refuge; neighbours ferried survivors in their own cars and a local businessman handed out two hundred kilograms of food — the last thing the rescuers asked for, he said, was baby food.

The country, meanwhile, performed its familiar rituals of shared grief: farmers' unions cancelled a planned tractor protest and stood in silence instead; striking doctors and teachers suspended their stoppages to help; regional governments from Asturias to the Basque Country offered beds and blood; and the chef José Andrés, watching from abroad, aimed a viral blast at the minister — Spain's trains, he wrote, were maintained worse than Ukraine's in wartime — that captured a mood no official statement could.

And the count began — led hour by hour by the regional president, Emiliano García-Page, who reached the field within the hour and whose "tragedy without palliatives" became the day's phrase, announcing the figures himself as they hardened:

— and the count kept changing all day, and the changing numbers themselves tell a story about chaos and hope:

- 10:00 — ten dead, announced by the regional government.
- Midday — thirty, then "more than fifty" estimated.
- 13:15 — 64: forty Iryo passengers, twenty-four AVE passengers, with six more bodies known to be inside a crushed carriage.
- 15:45 — 68, alongside the Prime Minister's visit: 44 Iryo, 24 AVE.
- The next morning, 29 October — 69, when a critically injured victim died in hospital. Sixty-eight of the dead were recovered from the wreckage; one died later of their wounds.

The injured counts wandered similarly: more than 180 by some totals, 110 admitted to hospitals in the region at one count, 31 still hospitalised days later, 13 of them in intensive care. Everyone aboard both trains — every one of the more than five hundred people — was hurt, bereaved, or both.

By the time the final number settled, Urda was the deadliest rail disaster in Spain in thirteen years, and the third deadliest since the Civil War era: El Cuervo 1972, with 86 dead; Angrois 2013, with 80; then Urda, 69. And it was the second catastrophic accident on Spain's high-speed network in nine months — because in January, at a place called Adamuz, forty-six people had already died on the same railway family. That earlier disaster is not background to this story. It is half of it.

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PART TWO — THE SYSTEMS, EXPLAINED OVER KITCHEN TABLE

Before we go further, your friend who knows nothing about railways deserves a plain explanation of the machinery everyone is talking about. Imagine the railway as a road with three layers of protection, and a fourth that was supposed to replace them all.

Layer one: the signals and ASFA. Since the late 1970s, Spanish trains carry a device called ASFA — Anuncio de Señales y Frenado Automático, "signal announcement and automatic braking." Think of it as a gadget that beeps at the driver when the trackside signals change, and can brake the train automatically if he ignores a stop. It reads little electronic "beacons" between the rails. ASFA is old, beacon-based, and limited: it gives information at specific points rather than continuously. A modern version, ASFA Digital, exists on most of the network.

Layer two: the LZB. For lines where trains go faster than 160 km/h (100 mph) (100 mph) — too fast for a driver to react to trackside signals — Spain's engineers chose, in 1992, the German LZB system. Picture a cable running between the rails the whole length of the line, crossed every hundred metres, whispering continuously to each train: here is your maximum speed, here is how far the line is clear, here is your target. Onboard, a display shows the driver three numbers — current speed, allowed speed, distance to the point where he must stop. If the driver exceeds the limit, the brakes apply themselves. In

LZB territory, the trackside signals are almost decorative: the rulebook says the LZB-equipped driver may ignore them. That is the system's genius and its fatal flaw — everything depends on the cable, the computers, and the centre.

Layer three: the rules for when it all breaks. The rulebook — the Reglamento de Circulación Ferroviaria, a public document — contains detailed procedures for a world where the LZB has died. The railway is divided into "cantons," stretches of track between signals that only one train may occupy at a time. When the electronic supervision fails, the system degrades to something called BSL — a block system where trains move on sight, watching the trackside signals, at reduced speed. The magic number is 40 km/h (25 mph), and the magic phrase is marcha a la vista, "running on sight": you move at a speed from which you can stop before anything you can see, because nobody is watching over your shoulder anymore. Before any train is released into this degraded world, the person in charge must personally confirm that the canton is free. The order that releases a train is called a telefonema, and it is read word by word over the recorded line.

There are even codes for how bad things are. A "Meta 40" message tells a train the track to the next station is free, with a 40 km/h (25 mph) cap. A "Meta 0" — sometimes written 000 — means the absolute opposite: occupied, forbidden to advance, danger.

Layer four: the future that never arrived. The European Union spent decades building a single standard system so that trains could cross borders without changing equipment: ERTMS,

the European Rail Traffic Management System. Its second level — ERTMS Level 2 — does what the LZB does, but digitally, by radio, and it is the mandatory future for every European high-speed line. Spain was a proud early adopter: as long ago as 2009, the state railway was promising fifteen hundred kilometres (about 930 miles) of it. But the very first high-speed line ever built in Spain — Madrid–Seville, opened by King Juan Carlos and Prime Minister Felipe González in April 1992 — was still running on its original LZB on the morning of 28 October 2026. It was, by then, the last high-speed line in the country that did.

Keep these four layers in mind: the beeping guardian (ASFA), the whispering cable (LZB), the rulebook for the dark (BSL and marcha a la vista), and the promised future (ERTMS). On the 28th of October, in the order we have seen, the whispering cable died; the rulebook for the dark was misread; the beeping guardian never had a chance, for a reason we will come to; and the promised future was sitting in the manufacturer's laboratory, installed on the line since June but not yet authorised for use.

One more technical point, because it decides everything. Remember the canton — the stretch of track that may hold only one train. When the LZB failed and both trains were halted, both trains were already inside the same canton. The track circuits knew a train was there — the control centre's own screen showed the section red, occupied — but the backup systems, the beacons and the balises, can only react when a train enters a canton. A train already inside generates no "entry" to detect. So

the ASFA had, quite literally, nothing to see. The green signal ahead of the stationary Iryo completed the deception: the only thing the approaching AVE's equipment could read told it all was well. As the newspaper elDiario's expert reconstruction put it: the majority thesis is that the AVE was already inside the Iryo's canton when the order to move was given.

How the safety architecture was supposed to work — the five barriers

Spanish investigators, following the European tradition of looking at systems rather than scapegoats, framed the accident as a chain in which five separate barriers failed one after another. Picture five lines of defence between a stopped train and a moving one:

1. **The LZB itself.** If it had kept working, the AVE's brakes would have been applied automatically, miles back, long before any human could have made a mistake. It failed at 08:30 — and in fact, in fairness, its failure worked as designed: it stopped everything. The catastrophe was not the failure but what came after.
2. **The BSL block discipline.** The rule that a canton must be verified free before any train moves. The two trains sat in the same canton, and the rear one was released anyway — with the control centre's own display showing the section occupied. An internal Adif instruction from January 2025, born from the recommendations of the previous accident investigation body, explicitly ordered: verify the canton free of trains first; never release the rear train until the front one

has moved.

3. **The recovery procedures.** With a Meta 0 — an absolute stop code — in force, the rulebook mandated marcha a la vista, 40 km/h (25 mph). What was dictated instead was "normal running," up to 200 km/h (124 mph). Another driver caught the error and had it corrected for his train; the correction never reached the AVE.
4. **The ASFA guardian.** It cannot brake a train for an obstacle inside a canton the train already occupies. Blind, through no fault of its own, at that point.
5. **The human control centre itself.** The last barrier is the room full of screens where all positions are, as the minister put it, "perfectly located at all times." The room received the warning at 08:48 and did not pass it on. Seventeen minutes had passed since the release order; there was, in the words of the minister, "more than enough time" to catch the error. Nobody did.

When the independent technical authority, AITAT, issued its first official note within twenty-four hours, its sentence became the case's epitaph: "a chain of failures, instructions and actions" had led to the collision, "without the necessary mitigating barriers having acted."

That is the machinery of the disaster, fully described. But machinery does not explain how it was allowed to exist. For that, we have to meet the people — the ones in the cab, the one in the control room, the ones in the boardrooms, and the ones in the courts.

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PART THREE — THE PEOPLE

The drivers

Antonio Acedo Meneses, the AVE driver, has already been introduced, and his story deserves to be told whole, because he is the one person in this tragedy whom the evidence — as it stands — most completely exonerates, without this document needing to exonerate anyone.

He was born in Móstoles, near Madrid, in 1990, into a railway family: his father was a train driver, his grandfather a Renfe electrician. He began his railway training at twenty-two, in 2012, later moved to Barcelona, and studied art, history and geography at the Open University of Catalonia alongside his driving. He was a photographer with a personal blog, and a member — indeed a transport-programme author — of a small party called Izquierda Española, whose leader, Guillermo del Valle, remembered him the day after the crash as "prudent, kind and exemplary," with a deep technical command of the sector. His party's remembrance said of him: "exceptional, good, prudent and committed… of those whose generosity leaves a mark." His own political programme — irony that needs no comment — diagnosed "the glaring lack of maintenance funding and the abandonment of all these years, translated into a demented deterioration of the railway infrastructure."

According to the evidence, he did everything right. He reported his train's fault correctly, using the correct code. He repeated the order back, as trained. And when the impossible appeared in front of him — too late, at closing speed — he pulled

the emergency brake and died in the collision. Emergency services, in the first hours, could not reach him on the radio. He never knew the full toll; he never had to. His colleagues at Renfe mourned him in a message: "Our colleague Antonio Acedo, driver of the AVE in the Toledo accident, has died. Antonio was an exceptional person, a good man…" His funeral, at the time of writing, has taken place quietly and without national coverage.

The **Iryo driver** — whose name, in contrast, has never been published, and this document will not be the one to publish it — survived. She was hospitalised, tested negative for alcohol and drugs, and gave a statement to the Guardia Civil. Her record that morning is, if anything, the most haunting of all: she stopped correctly, reported correctly, twice tried to restart her equipment, twice called for help, was told "whenever you can, alright?", and was then hit at speed from behind while sitting still, doing everything right, in a train full of strangers. She was still in communication with the control centre when the AVE struck. Her train's only failing was one she could not fix: for reasons its manufacturer has never explained, the Italian-built Frecciarossa takes longer to restart LZB equipment than Spanish-built trains. Its onboard software was designed for ERTMS Level 2 and Italy's own signalling system; LZB is a foreign layer for this fleet — a plausible engineering explanation that nobody has, on the record, actually confirmed.

There is one small mystery about the AVE's cab that deserves honest documentation: the drivers' union, SEMAF, in its first statement, referred to "the drivers who were in the cab" of the Renfe train — plural. Spanish AVEs normally run with a single

driver. Whether a second driver or an inspector was riding the cab that morning is, at the time of writing, unresolved in the public record, and is the kind of detail the judicial investigation will settle.

The operator

Of the control-room operator, we know only what the record permits, and it is important to say plainly: this person has been named publicly by the minister as the author of the error, has been interrogated by the police as a formal suspect — "investigado," not detained — attended with a lawyer, and exercised the right to silence.

The minister let it be known that the operator was experienced, that an auxiliary was placed at their side as the situation developed, and that a supervisor was also present. Three pairs of eyes, in other words, and the procedure still broke. Whether the error was one person's, the room's, or the system's is precisely what a court must decide, and this document will not pre-decide it.

What can be documented is the environment: a room that had been handling trains for two hours in degraded mode, with six trains halted, with garbled numbers on the radio, with staff who would later tell journalists that the centre has been chronically short of people, with no generational relay, with young staff pushed early into command positions. The union CSIF says it warned, in writing, in September 2026 — seven weeks before Urda — about excessive working hours in Adif's operation. The operator's own colleagues, through the Sindicato Ferroviario,

called the minister's instant "human error" verdict, hours after the crash, "precipitate," "bold" and "irresponsible." A union spokesman put the counter-frame succinctly: "sometimes human failures are due to how the organisation works." A railway engineer named Javier Matiacci had already said it in one sentence: human error is always a symptom, never a cause. For the record on those humans' conditions: a Renfe driver starts at a base salary in the low twenty-thousand euros, and even a senior AVE driver tops out, on the official published scales, below €57,000 a year before allowances — the wage context for the union's complaint about hours, and for the industry's struggle to hire (its biggest-ever intake, five hundred and fifty trainee drivers, had been announced for March 2026).

The investigators and the judge

The body investigating the technical causes is the **AITAT** — the Independent Authority for the Technical Investigation of Accidents, Spain's new multimodal accident investigator, which has existed in its current form only since July 2026, when it absorbed the old rail, maritime and aviation commissions. Its president, **Gonzalo Julián Guillén Espejo-Saavedra**, is a naval engineer and former sea captain — by training a maritime officer, now leading the investigation of Spain's worst rail disaster in thirteen years. Its council had six members at the time, with one seat vacant. The agency's whole investigative muscle for the crash amounted to three field investigators, later reinforced by two more — a staffing reality that newspapers contrasted with the scale of the task. The precedent, however, exists: after the Adamuz crash in January, the old commission invited the European Union Agency for Railways to observe its work, as EU law allows. Whether AITAT will do the same for Urda is an open and, many experts say, important question.

The criminal investigation sits with an ordinary provincial judge: **Aristea Afroditi Ortiz Tiliakou**, of Court Number 2 in Orgaz — a duty appointment. She is twenty-nine years old, by some accounts, and had taken up her post on 3 February 2026, as part of the 121-judge class that entered the judiciary that month. Her experience on the bench when the nation's biggest disaster case landed on her desk: eight months and twenty-five days. She is backed by the full emergency machinery: the regional high court's governing chamber activated the national

catastrophe protocol; Orgaz's number-one judge took the support role; and the number-two judge of Alcázar de San Juan — a member of the national network of emergency-management expert magistrates — became coordinating judge. Before anyone rushes to judgment about her, the record should note the parallel: the judge handling the Adamuz investigation, too, was freshly trained. Spain's system assigns its most enormous cases to its most junior judges, in the ordinary courts of small towns, and that fact — neither praise nor blame, simply fact — belongs in this document. Judge Ortiz Tiliakou's first orders have been methodical: seal the scene within a two-kilometre (a-mile-and-a-quarter) radius, seize the black boxes, ban all images of the victims, demand Adif identify everyone who held command that morning, place the proceedings under judicial secrecy — in the Spanish system's term, secreto de sumario — and order the specialist salvage firm Actren to work its crews in shifts around the clock, so that no night passes without the search for remains continuing.

The minister and the presidents

Óscar Puente, the Minister of Transport, learned of the crash while speaking at the opening of a world infrastructure forum in Madrid; he left immediately and spent the day at the Atocha control centre. His first press conference, three hours after the impact, set the narrative that would dominate the week: the LZB had fallen, the order given was "erroneous by any standard," and an operator had made it. "I am not thinking about myself," he said, asked about resignation; his emotional state, he added, was "irrelevant." His public record on rail safety is long and, at this point in the story, best summarised by its own words, documented later in this document: seven reprobation votes in parliament and four in the Senate over his handling of the Adamuz disaster; a criminal complaint admitted for processing against him (and re-routed on competence grounds) over that same crash; a public feud with the private operators about "parasiting" Renfe's workshops; and a documented track record of blaming operators first and systems later. He is, at the time of writing, still minister, and his government — which must call elections on 29 November — has rejected his resignation "outright."

The presidents of the two companies are a study in contrasts. **Pedro Luis Marco de la Peña**, president of Adif — the infrastructure manager whose centre gave the fatal order — was appointed in September 2024 after his predecessor was abruptly dismissed. A construction-industry man from the Basque Country, he stood beside the minister at the press conference

and said nothing. His own signature had meanwhile reorganised Adif's directorates from nine to six and installed the new operational chiefs — one a career Adif engineer doubling as head of the high-speed arm, the other a veteran of the liberalisation process taking over operations across some fifteen thousand kilometres (about 9,300 miles) of line. As this document was being finalised, the press reported that the government has decided his exit and is timing it. **Álvaro Fernández Heredia**, president of Renfe, was the minister's own city-bus manager in Valladolid when Puente was mayor — a connection the minister has dismissed with the technically-true observation that Heredia "is not even a member of the PSOE" (he is a former Más Madrid councillor). Heredia, who personally visited the Adamuz site in January, went entirely silent after Urda, speaking only through corporate statements. It was not his only register: that January he had faced a room of furious Catalan drivers in Barcelona after Gélida, and in April he had told the Senate's rail commission — while admitting rolling-stock failures — that he was open to merging Renfe and Adif under a single command. His silence after Urda, in that light, was at least one newspaper's story of its own.

That newspaper's investigation, published in Galicia and read across the country in the crash's first week, documented what it called the replacement of the network's technical leadership by political appointees — naming, among others: the Renfe operations director José Alfonso Gálvez, who had been Heredia's traffic manager at the Valladolid bus company, admitted to his post while still a ministry official, the sole successful candidate

of twenty-eight; the minister's former land-transport secretary Marta Serrano, given a Renfe directorship abolished seven months later; the former party colleague Ismael Bosch, placed on the Renfe Viajeros board; the general-secretary post, taken by José María Vicent, whose twenty-five-year professional requirement was scrapped; the Basque party executive Carlos María Juárez, made director-general of the rail sector after the career-civil-servant requirement was likewise scrapped, behind a vacancy voided and readvertised with narrower criteria; the crisis-communications consultant Carles Montaña, with no railway experience, made operational emergency director; and the circle of appointments from the Valencia political world — Pere Rostoll at the ministry's communications, Salvador Almenar in the Adif president's private office, Roser Obrer, Joan Calabuig, and the Rodalies chief Josep Enric García Alemany, sacked that January — rounded out by the bus-industry arrivals Rafael Cortés and Antonio García Pastor, both of Avanza, the latter the only candidate of seventeen.

The exposé's unnamed sector sources supplied its thesis: in public administration, one said, leadership requires humility and respect for teams — replace that with recklessness, and you run serious risks. The minister's defenders answer that the appointees are competent managers; the document records both the taxonomy and the defence, and leaves the judgment to the reader. Iryo's leadership, by contrast, was vocal: its CEO, **Fabrizio Favara**, had warned the Senate about the LZB exactly three weeks before the crash; its president, **Carlos Bertomeu**, said the unsayable the day after: "this should never have

happened."

The victims — the faces of the 69

Behind every number in this document there was a person with a morning plan. Sixty-nine people left home on the 28th of October expecting an ordinary Tuesday; their plans included a job interview in Seville, a commercial shoot, a scientific congress, a visit to a bull-breeding farm, and a niece's first weeks of life. The identification work — grim, slow, DNA-based, carried out by forensic teams in Toledo with families waiting in a care centre staffed by twenty-seven psychologists — had confirmed fifty-six of the sixty-nine by the end of the first weekend. The names that follow are the ones confirmed in the public record at the time of writing; the list is not yet complete, and its incompleteness is itself part of the story.

From the **Iryo**: **Fátima Báñez**, 59, former Minister of Employment under Prime Minister Rajoy, later president of the CEOE foundation, travelling home to Huelva. **Marisa Manchado**, 70, a composer who had won Spain's National Music Prize in 2024. **Antonio Vázquez**, 75, brother of the bullfighting maestro Curro Vázquez. **Ana Martina Conte Grand**, 32, an Argentine podiatrist who had recently moved to Spain, on a family visit. **Patricia Fernández Häring**, an architect active in Madrid's architects' association. **José Ángel San Emeterio Freire**, 54, a brother of the Seville brotherhood of the Museo — his dog Trece, travelling in a carrier, was found alive by the Guardia Civil and returned to his widow.

From the **AVE**: **José Manuel Chacón**, 31, a Guardia Civil officer from Seville; **Juan José García Palacios**, 26, another Guardia Civil officer. **Raúl Triguero**, 43, a television reporter heading to an interview. **Julieta Gallego Gámez**, 24, a long jumper who had turned twenty-four nineteen days before. **David Reina**, 22, a Renfe technician himself, from Málaga. **Iván Romano Ramos**, 42, a national police officer. **Adriana Marroquín Rodríguez**, 27, an industrial engineer. **Mercedes Irazola Bezares**, 40, from La Rioja. **Mari Carmen García**, 26, of Mallorca, travelling to film a commercial. **David Jiménez Gil**, a biochemist and microscopy-industry director. **Carlos Ortiz**, a schools director. **María del Carmen Vázquez Nieto**, 42, an architect from Galicia. **César Vicente Sanz**, father of an actor. **Tamara González** and **Marta Shahbazi**, two scientists on their way to a biology congress in Huelva. **Cristina Ferris Villanueva**, 40, a pharmacist at the Spanish medicines agency. **Antonio Montero Domínguez**, a tax inspector. **David Martínez Méndez**, 25. **Francisco José Barbero Liébanas**, 32. **Agustín Randazzo**, 51, an Argentine telecoms professional. **Andrea Olmos**, a visual merchandiser from Figueres — and travelling with the Argentine podiatrist named above, her brother, who survived with injuries. A Portuguese citizen, unnamed in the press. And **María de los Ángeles M.**, a hostel owner from Cuevas de San Marcos, the one who died in hospital, the sixty-ninth.

Also among the dead, from the record: a **Mexican national** reported missing by his family (who gave DNA samples), and a **Czech citizen**, confirmed by Czech authorities. Fifty-one

foreign nationals of twenty-three nationalities were aboard the two trains; two of the dead were foreign — one French, one Portuguese, at the first confirmations.

Fifty-six identified, thirteen still being named as this document closes. Behind each name on this list there is a family that gave DNA and waited. That wait is part of the story too.

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PART FOUR — THE MONEY

Now we follow the money, as this document's method requires — because the LZB that failed in 2026 was not supposed to exist anymore. The story of why it still did is a thirty-year story of contracts, collapses, cartels, and deadlines that came and went. Every step below is documented in official records: a regulator's ruling, a ministry press release, an official journal, a company announcement, or the European institutions' own documents.

1992: The line is born, and so is the LZB

When King Juan Carlos and Prime Minister Felipe González inaugurated the Madrid–Seville line in April 1992, it was the showpiece of a modernising Spain: 470 kilometres (about 292 miles) of new railway, twenty-first-century trains, and German continuous-cab-signalling technology — the LZB — that let them run at 300 km/h (186 mph) safely. The signalling on that line was supplied and maintained, from its first day, by a company called Thales — or its French ancestors in the business. As Thales itself would proudly note three decades later, its relationship with the line went back "much earlier than 1992. Since we built it, more than thirty years ago."

For the next fifteen years, that arrangement was stable, comfortable, and — as a regulator would later rule — criminal.

The cartel years: 2003–2019

In July 2019, Spain's competition regulator, the CNMC, issued one of the largest rulings in its history: eight companies fined a combined €128 million for running a bid-rigging cartel in Adif's signalling tenders worth €4.1 billion over fifteen years. The two cores of the cartel were Siemens Rail Automation and Thales España — the same two firms that dominated the market. The ruling's annexes read like a crime novel's wiretaps: internal emails from 2013 and 2014 in which executives divided the country between them with what they called "the historic agreement" — Madrid–Seville to Thales, Madrid–Valladolid to Thales, Madrid–Levante to be discussed, maintenance contracts to be covered for each other. A handwritten note from January 2015 by the company's projects director — Fernando Ortega Díaz, whose career would later place him at the head of Thales España's transport division precisely through the years of the renewal — proposed "covering for Siemens Rail by subcontracting Thales in the Madrid–Valladolid maintenance contract."

One email, from December 2014, deserves to be quoted at length because of what came later. Discussing the coming renovation of the Madrid–Seville line, a Thales executive wrote that the historic agreement should be respected: the new European signalling for Madrid–Córdoba would be Thales's, for Córdoba–Seville Siemens's, "all the new interlockings would be Thales's, and all the maintenance (20 years) for Thales. Track circuits for Siemens (what a pity ours isn't certified yet)."

Six years later, the renovation of Madrid–Seville was indeed awarded. To Thales.

The regulator fined Thales €17 million and Siemens Rail €18.9 million, and named executives on both sides. It also recorded, with some internal discomfort, that its own transport directorate had once considered the firms' cooperation technically justified. The firms' lawyers disagreed with everything. The fines were paid. And in October 2021 — in the very weeks the cartel fines were making news — the relicitation of the Madrid–Seville safety systems reached its conclusion. The award went to Thales España, at 28.45 per cent below the budgeted price. One conservative newspaper called the winning bid "rash"; another would later note that the price was so low the European Union accepted the companies' explanation for it.

The 2020 tender that collapsed over a mistake

The path to that award had already eaten a year, and the cause of the delay is one of the strangest documents in this whole story. The tender for the renewal had been authorised in early 2020: €108 million, fifty-two months, and at its heart the replacement of the LZB with the modern European system, ERTMS Level 2. Siemens — the same Siemens that had challenged nothing in fifteen years — filed a legal appeal. And the appeal revealed something remarkable: Adif's own technical staff, answering bidders' questions, had twice told the companies that alternatives which "dispensed with LZB" once ERTMS entered service would be considered. But the tender documents themselves required the exact opposite: the LZB had to be kept, guaranteed, wired into the new interlockings throughout the works. Some bids were built on the erroneous answer, others on the official requirement, and the regulator's tribunal agreed the situation was incurable — the principle that all bidders compete on equal terms was broken. Rather than risk a voided award, Adif abandoned the entire tender in November 2020, a year after launching it. The relicitation was published days later. The lost year pushed the project's horizon from 2024 into the second half of the decade — and, as it turned out, beyond.

There is a detail in that ruling worth pausing on, because it is the hinge of the whole tragedy. The contract that was supposed to retire the LZB legally required the LZB to be kept in commercial operation "before, during and after" the new

system's commissioning. The migration was designed as a long overlap, years long, in which the old system would carry the traffic while the new one was built alongside it. Everyone signed that clause. It means the crash of October 2026 did not happen because a plan had gone wrong. It happened inside a plan working exactly as written — a plan whose own terms guaranteed that the 1992 system would still be the operative one on the morning of the 28th.

2021: The award, and the same week, a fine

On 16 September 2021, Adif's high-speed arm awarded the contracts: €136.4 million in total, EU-financed. The centrepiece — roughly €94 million — went to Thales España for the new interlockings and ERTMS Level 2 across the whole line, with a fifty-two-month execution calendar running to February 2026. A further €21.5 million went to Thales for the track circuits; a consortium of Ferrovial, Azvi and others took a €21 million slice for the points and crossings. The maintenance of the line — LZB included — was also Thales's, renewed in a separate €61.5 million, three-year contract in August 2026, two months before the crash: the same company that built the 1992 system had kept it alive for thirty-four years and was still its keeper on the morning it fell.

One more corporate twist belongs in the record. In the same month the contract was awarded — September 2021 — Thales announced it was selling its entire ground-transport business to Hitachi. The deal took three years to close, completing on 31 May 2024: Hitachi Rail acquired the business for €1.66 billion,

inheriting the Madrid–Seville renewal mid-delay. So by the time the contract missed its first deadline, the signatory of the 2021 award no longer owned the work; a new corporate master held a contract already years behind — and would install a new chief executive of its own, Giuseppe Marino, in April 2026, six months before the crash. (The Spanish subsidiary, for its part, is led locally by a president named Pepi Andrea.) When journalists asked Hitachi Rail why the migration was unfinished and what the June 2026 extension changed, the company declined to explain.

The deadlines that came and went

The official record of the promised dates now reads like a catalogue of hope:

- **2024.** The original tender, before its collapse, aimed for ERTMS by now.
- **April 2025.** The Ministry of Transport's own blog, celebrating the line's 33rd anniversary, announced it was in the "final phase with 97% of the investment executed or in advanced execution," promising the line "completely renovated" with ERTMS Level 2 by its 34th anniversary. The same post conceded ERTMS Level 2 was "the element requiring the longest execution period."
- **"Beginning of 2026,"** per Adif's website as recently as June 2024.
- **February 2026,** per the contract's own fifty-two-month calendar.

- **April 2026,** per the ministry's later public commitments — the 34th anniversary it had promised.
- The deadline passed. On 18 January 2026, two days after the contractual completion date, the Adamuz disaster killed forty-six people. That is a coincidence of dates, but it is a true one, and this document records it as such.

By the summer of 2026, Adif faced a choice about the overdue contract: terminate it, extend it with penalties, or extend it without. Records show it chose the third. In June 2026, the contract was extended on unchanged conditions, with no penalties applied, pushing the expected completion into 2027. The same summer, Spain's Recovery Plan was being renegotiated with Brussels for the ninth time in three years — milestone tables softened in 2024 and again in the amendments of January and June 2026, precisely so that objectives like these could be redefined rather than failed. The EU's own bank, in the project document that accompanied its €90 million loan for the renewal — signed in March 2022 — had described the line's installations as being "at the end of their useful life." Brussels, diagnosing the line obsolete years earlier, had already granted €111.6 million toward its modernisation.

And so we arrive at the last public word on the schedule, spoken by the minister himself in the first hours after the crash: the ERTMS had been installed on the line in June 2026, was in laboratory testing, and would be operational "probably in January 2027."

On the morning of 28 October 2026, the trains of Spain's first high-speed line ran on a system that the European Union's bank

had declared at the end of its life, that four governments had promised to replace for six years, that had been contractually guaranteed to survive through any migration, and that its own manufacturer admitted was no longer even built. "We're replacing it," the minister had explained months earlier, "not because it isn't good, but because it's no longer manufactured."

What the money bought — and what it didn't

A word, here, about what the renewal money did accomplish, because an honest document must not leave the impression of simple theft or pure neglect. The Madrid–Seville line did receive a genuine modernisation: nine Sierra Morena viaducts renewed; over a hundred high-speed points replaced; nearly two hundred thousand sleepers renewed or in renewal; electrification upgraded; new object-detection cabinets; a doubling of GSM-R coverage. The government counted the programme at €680 million in 2023 and €780 million by 2026 — and Spain's engineers, reviewing the figures after the crash, added the essential caveat: money awarded is not money executed, and equipment installed is not equipment authorised. The only number that mattered on the 28th of October was that the one system carrying the trains was thirty-four years old, failing at a rate of roughly one hundred and seventy times a year — a frequency the minister called "within the normal parameters in Europe," with about three-quarters of the falls attributed to causes outside Adif's control, from cable theft to storms — and that the morning's failure was, as the incident report would show, far more than a single fall.

Whether any of this amounts to criminal responsibility is a question for the courts — and the courts are already asking it. After the Adamuz disaster, the European Public Prosecutor's Office opened an investigation into the possible embezzlement of EU funds in the maintenance of that stretch of line; a judge, José Luis Calama, found that Adif had removed forty-two metres of fractured rail from the scene without judicial authorisation (another thirty-six metres were recovered from a company depot), and the prosecutor demanded every contract file for the stretch. That investigation, like the minister's own reprobations and the criminal complaint bearing his name, predates Urda. The money trail, in Spain's railway disaster story, never stays cold for long.

One last commercial fact completes the picture, because it is the reason the corridor was so full that morning. Four years of rail liberalisation — Iryo and Ouigo competing with Renfe since 2021 — had doubled traffic on the Madrid–Barcelona axis and cut fares on it by two-thirds, making Spain the second-largest market in Europe for private high-speed operators. The infrastructure charge that paid for the network had itself been the subject of a regulator ruling and a ministerial feud in the months before the crash, the minister accusing the private operators of "parasiting" Renfe's workshops while they complained of maintenance delays that had tripled. The corridor's boom and the corridor's obsolescence grew together, side by side, for four years.

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PART FIVE — THE WARNINGS

A disaster like this one is rarely a surprise in retrospect. What investigators call the "swiss cheese" model says that behind every accident there are people who saw the holes lining up. This part documents every warning that is on the public record — who gave it, when, and to whom — because the pattern of what was said and what was heard is as much a part of this story as the crash itself.

September 2023: Brussels. The European Commission flagged the obsolescence of the Madrid–Seville line and granted €111.6 million toward its modernisation. The EU's bank, in the project document that accompanied its loan, had written that the line's installations were at the end of their useful life.

August 2025: the drivers' union writes a letter. SEMAF wrote formally to Adif about the state of the line — warning of track wear on the corridor and of a "profound and accelerated degradation" in the rolling stock. An important precision, verified by the fact-checkers at Maldita: the letter asked for speed reductions on stretches limited for engineering reasons, and the Adamuz stretch specifically was already limited to 250 km/h (155 mph); so the letter did not, strictly, predict Adamuz. But its rolling-stock warning was general, and its tone was alarm. A year and three months later, its author would call a sector-wide strike and be told by the minister that the drivers were simply in a bad "state of mind."

January 2026: Adamuz. After forty-six deaths at Adamuz and one at Gélida, the union demanded criminal accountability

of safety managers and announced a strike across the whole sector. The government's response, on 10 February, was a money-and-jobs agreement: €1.8 billion more for maintenance over the coming years, 3,650 new jobs, 2,400 Adif posts by 2030, three hundred new maintenance vehicles. A real answer, on paper — eight and a half months before Urda.

Spring 2026: the safety regulator's circular. The AESF — the Spanish Railway Safety Agency, the state's own watchdog — circulated a recommendation after Adamuz: when systems fail, run at sight, at 40 km/h (25 mph); and the person in charge must verify the canton free before any movement. The circular described exactly the failure that would kill at Urda, and prescribed exactly the remedy that would not be applied.

September 2026: a regulator is fired, and warns. Rocío Báguena, a career civil engineer and head of the transport ministry's land-secretariat — who also chaired the AESF's council — was dismissed on 1 September 2026, at the minister's proposal, without explanation. ("The minister has given me no explanation for the dismissal," she told senators later, under oath.) Twenty days later she appeared before the Senate's investigative commission on the state of the rail network and said, in her professional opinion, that the LZB was "already obsolete," adding that she believed the replacement would not be complete "for a few more months." She also told the commission something almost nobody knew: that the safety agency had never been asked to authorise the Adamuz stretch for service — because Adif self-authorises sections during works and only requests the agency's service authorisation at the very

end, after thousands of trains have already passed. She had been the state's top transport-safety official until weeks before; the state was no longer listening to her.

September 2026: the private operators testify. On 24 September, Ouigo's director-general told the same Senate commission that the system was "a bit old" and that every expert said it had to be replaced — while carefully adding that her company had never detected a grave safety risk. On 5 October, Iryo's CEO Fabrizio Favara went further. The most frequent problem his drivers reported, he told the senators, was the LZB: when the connection between ground and train broke, "the train stops, cannot move, and takes quite some time to reconnect." His drivers had been reporting these incidents to Adif's traffic regulation centre since 2025. Three weeks later, an Iryo train sat immobilised by exactly that failure, in front of a red signal, while the state railway's fastest train bore down on it in the dark, ordered to "normal running." A worker from another operator — a driver on the line the night before — had stopped at Urda itself at half past eight the previous evening when his train "did not recognise the system." He was stopped for eleven minutes, at the same station, thirteen hours before the crash. And the union CSIF says that in September it formally warned about excessive working hours in Adif's operation.

All year: the incidents themselves. Adif's own public alert channel recorded at least ten signalling incidents around Urda since 2024 — in May 2024, July 2025, four in 2026 — the most recent in July, caused variously by cable theft, fires near the line, and equipment faults. On the morning itself, in the control

room's own recording, an operator says it out loud: "The Urda central has fallen… we're deciding whether to disconnect or not." And under the procedural chatter, a final, quiet pattern: the year 2026 would see at least 170 LZB failures on the line, which the minister, after the crash, described as normal.

Every one of these warnings is documented. Not one of them, on the record, produced the one change that mattered — a ban on releasing trains at "normal running" into a degraded line — before the 28th of October. After the crash, the drivers' unions demanded exactly that ban, as loudly as they could.

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PART SIX — ADAMUZ: THE FIRST CRASH OF 2026

You cannot understand Urda without Adamuz, because Adamuz is where every actor in this story first appeared — the minister, the companies, the regulator, the judge, the investigators, the union, and the rail itself. What follows is the full story of the evening of Sunday, 18 January 2026.

The evening

It was about 19:45 on a winter night, on the high-speed line between Córdoba and Málaga. An Iryo Frecciarossa, service 6189, was running north toward Madrid with about three hundred people aboard, at around 110 km/h (about 68 mph) through the station of Adamuz. Without warning, the last carriages of the eight-car train lurched: a rail beneath them had broken. The train derailed sideways, its tail spilling onto the adjacent track. The driver fought for control and stopped; her train sat there, tilted, blocking the other line.

Twenty seconds behind — too close, in railway terms — came an oncoming Renfe Alvia, service 2384 from Madrid toward Huelva, with 187 aboard, running at about 200 km/h (124 mph). Its driver had no time; even the chief of Renfe would say human error was "practically ruled out" — twenty seconds is not enough to see, decide, and brake a 400-tonne train. The Alvia struck the derailed cars head-on. Its first two carriages fell down a five-metre embankment. Most of the forty-six dead were in those front cars.

The drivers' black-box radio tells the story in real time, and it is worth hearing, because it will echo in your mind at Urda. 19:45:02 — the Iryo driver calls in: "I've just suffered a catch at the height of Adamuz." The controller: "Ah, yes, I can see you." She is ordered to lower her pantographs; she answers: "They can't be any lower. In fact, my train is locked." 19:49:33 — nearly five minutes later — she calls again: "It's a derailment. I am invading the adjacent track. I repeat, derailment." She begs: "I need you to stop the traffic on the tracks urgently, please." And the controller answers, in the words that would be replayed all winter — "Sí, sí, sí… no hay ningún tren llegando": yes, yes, yes, there is no train arriving. Within seconds, the Alvia hit.

She had asked for the line to be closed. It was not. Between the calls, the control centre had spoken with an AVE travelling behind the Iryo, and made two unanswered calls to the doomed Alvia's driver. The crash, investigators would later establish, happened in under nine seconds — not the twenty originally reported — in a stretch that was, in the driver's words, "completely dark." The president of the accident commission would call it "relatively normal" that a controller might not notice an impact in the dark. The drivers' union called it something else.

The rail

The physical evidence, assembled by the commission and the Guardia Civil, told a story of metal fatigue older than some of the victims. The broken rail was a weld joining two generations of steel: an original rail rolled in 1989 and laid in 1992, and a new rail installed in a 2023 renovation. The fracture had begun more than twenty hours before the accident, in a weld joining two generations of steel — an ENSIDESA rail rolled in 1989, inscribed "89," laid in 1992, welded to a modern "ENSIDESA 23" rail installed in the 2023 renovation — and the paperwork then produced a documented contradiction: the renovation contract had ordered such welds made with the higher-grade steel's kit, while Adif's own internal norm orders the opposite. The minister attributed one of the two documents to "an error." A small crack, invisible to the line's ultrasonic inspection trains, grew with every passing train. Up to twenty-five trains ran over that growing crack in what a senator would later call "Russian roulette." Three trains had passed just before the derailment, all with wheel damage noted afterwards; the maintenance database recorded track problems on the stretch from 2022 onward. The line had been renovated in May 2025 at a cost of €52.5 million, the work executed by a consortium of Ferrovial, OHLA, FCC and Azvi — and the fatal weld, it emerged, had ultimately been done by a small Galician subcontractor, Maquisaba, that almost nobody had heard of, supervised by an engineering firm, within an audit regime that the safety agency itself had flagged as weakened by subcontracting chains.

Then came the scandal inside the scandal. A judge, Cristina Pastor — newly trained, like her Orgaz colleague of October — inspected the site on 17 March 2026 and found that Adif had removed forty-two metres of the fractured rail without judicial authorisation, to "restore service." Another thirty-six metres were discovered at a company depot. The judge warned the company formally; the rail was sealed and sent for testing. The European Public Prosecutor's Office opened an investigation into the possible embezzlement of the EU funds — over €100 million — that had paid for the stretch's maintenance. The central forensic question became whether money meant for the track had actually reached it.

The engineering argument that followed deserves to be documented in full, because it is the same argument that will follow Urda. The renovation contract's technical specification had ordered welds between the two rail grades to be made with the higher-grade steel's welding kit; Adif's own internal norm ordered the opposite — the lower grade's kit. One of these documents was wrong, and the minister attributed it to "an error." The unions asked the judge to summon as formal suspects every technician in the welding chain. And underneath it all lay an older ghost: a 2016 report by the same accident commission, on the Pajares tunnel derailment, had urgently recommended that the track-inspection regime be strengthened so that its ultrasonic trains could "effectively detect" defects before they became breaks. Eight years later, a crack grew under twenty-five trains undetected, and the question of whether those recommendations were ever implemented — in

the words of the press, it "remains in the air."

The political season

Adamuz set the political weather for the rest of the year, and everything that happened then reads today like a rehearsal for October.

The minister, Óscar Puente, initially said the stretch had been "completely renovated" — and when journalists revealed that the renovation had mixed new materials with 1989 components, the opposition spent the winter on one word: "He lied." By mid-February, parliament had voted to reproach him seven times, and the Senate four. On 29 January, in a Senate appearance, he defended himself for over two hours — "neither has information been lied about nor hidden"; rail breaks were "habitual"; an integral renewal did not mean "lifting it entirely and rebuilding it." The People's Party senator's reply entered the lexicon of the crisis: "The heart of the scandal: lying to a country while the wounded were being tended and the dead recovered… The hoax is you." A Vox senator told him his "corruption, unfortunately, kills." The minister's answer was memorable: "Do you know why they ask for my resignation so much?… Because I do it very well."

A criminal complaint against him — for reckless homicide, reckless injury, and abuse of office — was registered in the Audiencia Nacional in late January; the prosecution ruled the court not competent, since a minister belongs before the Supreme Court, and the matter was rerouted. The Supreme Court trial of the previous era's corruption — the "masks case"

that had ensnared the former minister José Luis Ábalos and, in its papers, Adif's former president Isabel Pardo de Vera — pandemic masks, an Adif contract of €12.5 million, police findings that the former president's devices had been wiped of messages from the relevant weeks — unfolded that same April in the Supreme Court, sending Ábalos and his fixer to prison, with eighty witnesses including the current president of Adif. The case had already cost Adif's personnel chief his job in 2024, and a private prosecution had asked that the transport minister be called as a witness; the judge declined, for then, leaving the door open. The same winter, the opposition tabled sixty-five parliamentary questions on technical rail-safety issues and demanded the files of contracts on the affected stretches — officially citing their possible links to people implicated in corruption cases. A newspaper tally would later count fifteen parliamentary requests for explanations on rail accidents since 2023; the official record shows every one closed without agreement or decision. And the victims' home province of Huelva — twenty-eight of the forty-six dead came from there — marched five thousand strong in March demanding "the whole truth." The state paid homage in Huelva on 31 January.

And the technical conclusion, when it came in the spring, was almost an anticlimax in its narrowness: a rail had broken, probably at a defective weld. The final report was still pending; new findings were possible.

What Adamuz explains about Urda

Keep Adamuz in mind as we proceed, because every element of Urda existed already in January: a degraded-mode corridor supervised by the same centre; a control room that heard a driver's urgent call and answered "there is no train arriving"; an investigation handed to a rookie judge in a small town; a minister blaming individuals first; a rail industry whose contracting chains ran from the biggest construction firms to an anonymous subcontractor; a regulator that learned of its own network's risks second-hand; and a public that was told, in essence, that the system was safe. When, nine months later, the same minister stood before the cameras with a new disaster and a new single culprit — "an operator" — Spain did not need to imagine what came next. It had watched the film before.

* * *

PART SEVEN — THE YEAR OF DISASTERS: 2024–2026

Zoom out once more, and Urda stops looking like a single accident and starts looking like the largest event in a pattern. Here is the pattern, year by year, as documented.

October 2024: the tunnel train. On 19 October 2024, a Renfe service train broke down on the steep ramp into Madrid's Chamartín tunnel and began to roll backwards, driverless, with two technicians aboard and no brakes, gathering speed through the high-speed tunnel toward the capital. An Adif employee at a trackside post made a decision that probably saved lives and certainly entered legend: he deliberately routed the runaway onto a diverging track, derailing it against the tunnel wall. Three carriages overturned; nobody died; damage exceeded €2 million. The minister confirmed the deliberate derailment. The lesson of Chamartín is not the heroism — it is that when the technology failed, the system was saved by one human's improvisation, exactly the kind of improvisation the rulebook tries to replace with procedure.

The autumn cluster. Within twelve days that October: the tunnel runaway; a dangerous-goods derailment at Matillas; a rockfall that derailed an Alvia with 291 passengers near Álora. In 2023, a Talgo derailed leaving Atocha on a broken switch blade; in October 2025, a commuter train's last car derailed on a bridge over the Jarama, its cab hanging over the parapet; the derailment statistics had been rising for a decade. And the conventional network — the old lines the high-speed system left

behind — spent 2026 in open crisis: lines closed for a year in the east, one line never reopened since a 2021 storm, another delayed by eagle nests, public disputes between Renfe and Adif over who ordered what closure. The pattern section of any honest investigation file begins to write itself.

20 January 2026: Gélida. Two days after Adamuz, on the night of a major storm, a retaining wall beside a motorway above the railway near Barcelona collapsed onto the tracks in the rain. A commuter train ploughed into the rubble at speed; the driver, a 28-year-old trainee from Seville with two fellow students in the cab, died. All thirty-seven aboard were injured. The entire Catalan commuter network was suspended for inspection. The wall's ownership — road authority or railway — remained, for months, unassigned.

12 October 2026: Ricla. Sixteen days before Urda, an AVE on the Madrid–Barcelona line caught fire at Ricla, in Zaragoza; 283 passengers were evacuated on the track, several treated for dizziness. No one died. The corridor was interrupted for days.

28 October 2026: Urda. Sixty-nine dead.

The tally of 2026's railway deaths — forty-six at Adamuz, one at Gélida, sixty-nine at Urda — passes one hundred and sixteen. It is, by any measure, the worst year in Spanish rail safety since the 1970s, and it has happened on the watch of a minister who had already survived seven parliamentary censures over the first of its disasters. That is not an accusation. It is the arithmetic of the record.

Now to the institutions that were supposed to see the pattern — and to the strangest and most revealing part of this whole

document: what happened to the truth about the crash, in the hours and days after it, online.

* * *

PART EIGHT — THE INVESTIGATORS: WHO WATCHES THE WATCHMEN

Every disaster of this scale gets two investigations: a technical one, which asks "how did the system fail," and a criminal one, which asks "who, if anyone, is responsible." Spain's machinery for the first had just been rebuilt; its machinery for the second is ordinary provincial justice. Both deserve a close look, because the health of the answers depends on the health of the askers.

From CIAF to AITAT: a watchdog rebuilt mid-crisis

For two decades, Spain's rail accidents were investigated by the CIAF — a small commission attached to the transport ministry, which everyone agreed was better than nothing and few agreed was independent: its members were drawn largely from the ranks of the very companies whose accidents it investigated, its budget was modest, its reports slow, and the European Commission had formally complained in 2019 that its methods breached EU law. Victims' associations had spent a decade saying so; one filed a complaint in 2014 stating flatly that the commission "does not comply with European regulations" and "is not independent."

The reform was written into law in August 2024: a single independent authority covering rail, sea and air, on the European model. The political argument over it is worth recording because it previews every argument since: the governing parties voted it through; the People's Party voted

against; Vox abstained. The statute took until February 2026 to finalise, and the old commission was not formally extinguished until July 2026 — which means the CIAF investigated Adamuz and the new AITAT, barely three months old, with a council still missing one member, investigated Urda. The old commission's last composition, exposed by the press in the Adamuz weeks, matched its reputation: six of its seven members were former senior officials of Renfe or Adif.

The new body's record so far, documented in its own first official note on Urda, is professional in method and striking in scale: three investigators on the ground within hours, the interlocking records seized at Urda, the control centre's order book and videographics copied, the black boxes recovered and locked in Guardia Civil custody pending the judge's conditions, the drivers and controllers not yet interviewed because they were, in the note's restrained phrase, "not in a condition" to be. The final report is due within a year. And the president who signed that note — remember — is a naval architect, doing the job of a railway investigator, with three field staff.

One more institutional fact belongs here, because it will matter for years: the safety regulator, the AESF, never authorised the Adamuz stretch for service at all. Adif, by procedure, self-authorises sections during works and applies for the agency's blessing only at the end. Iryos and Alvas ran for months at full speed on a stretch the state's own watchdog had never signed off. The woman who revealed this, Rocío Báguena, had run the ministry's land transport directorate and chaired the regulator's council — until her dismissal on 1 September 2026,

three weeks before Urda, for reasons the minister never gave her.

The criminal files

The judicial map of the whole affair, as this document closes, runs like this. The Urda case sits with Judge Ortiz Tiliakou in Orgaz, backed by two senior magistrates. Two private prosecutions have registered: one by the union Manos Limpias against the infrastructure manager, the operators and responsible individuals; one by Vox, which demanded — among other things — that Adif identify by name every person who held command in the Atocha centre between 7:00 and 11:00 that morning. The operator who gave the order was interrogated on 29 October as an "investigado" — a formal suspect, not detained — with a lawyer, and chose to remain silent. The Audiencia Nacional separately admitted a criminal complaint against the transport minister himself over Adamuz — filed by a private association, Libertad y Justicia, before judge Antonio Piña — alleging homicide, injury and abuse of office, before the court ruled itself not competent, since a minister answers to the Supreme Court; that complaint's fate, at this writing, is unresolved. And the European Public Prosecutor's fraud investigation over Adamuz's EU maintenance funds remains open, its files running from the €52.5 million renovation down through the subcontractor who held the welding kit.

Whether any of these files ever names a defendant beyond the operator is the question on which everything else now turns — and this document makes no prediction. It notes only what the

record shows: a rookie judge with her scene under seal; an accident authority with three investigators; a regulator that learned of its own network's risks from a dismissed director's Senate testimony; and a political system that has spent the year voting on the minister while the minister's ministry renegotiated the deadlines the crash would expose.

* * *

PART NINE — THE INFORMATION WAR

Within hours of the crash, a second battle began: the battle over what the crash meant. Its weapons were leaked audio, synchronised headlines, official framing, and — within a day — outright fabrication. Every part of it is documented, and the documentation itself becomes evidence about how modern disasters are explained.

The leak and its choreography

At nine o'clock that evening — roughly nine hours after the impact, and a few hours after the minister's press conference had planted its narrative in a single word, "error" — three of Spain's major news organisations published, simultaneously, recordings and transcripts of the control room's communications: the SER radio network (Guillermo Lerma), the newspaper El País (Carlos E. Cue), and the online daily elDiario.es (Esther Palomera) — each with its own byline and each with the same material. The recordings showed what you have already read in this document: the "normal running" order, the driver's readback, the Iryo driver's two calls, the chatter of a confused room.

The choreography did not go unnoticed. Conservative outlets immediately asked who had handed three friendly newsrooms the state's own evidence within hours of the disaster, and answered their own question: the government, they wrote, had "filtered" the audio "to allied media to blame Adif technicians" — a firewall, one analysis called it, drawing a circle around a single

operator and keeping the focus away from the system, the obsolescence, the missed migration, and the political chain above the control room. The leak outlets' defenders replied that the recordings were genuine and their publication legitimate journalism — which they were. Both things are true at once, and an honest document holds them together: the audio was real, its release was lawful journalism, and its synchronised timing, hours after the minister's framing, shaped the first day of public understanding toward a single pair of hands. One small anomaly belongs in the record beside it: a broadcaster reported that the accident authority's account of the failure chain vanished from the international version of its own first document — a discrepancy the authority has not explained, and which this document records as an open question rather than a conclusion. The editorial packages made the frame explicit: one front page read "one technical failure and three human errors." The union CSIF called the minister's instant verdict a use of a worker as a "scapegoat," and demanded his resignation — its own, earlier warning about working hours suddenly very relevant.

The fakes — and one true video from April

Then came the fabricated layer, and here the speech of this document's title returns to the story.

Within a day of the crash, social networks were circulating a video of Israel's prime minister, Benjamin Netanyahu, speaking to camera in April — with new subtitles claiming he had threatened Spain with "sudden power cuts" and "train accidents." A second image, showing the prime minister at a

whiteboard with "Spain" written on it, turned out to be AI-modified. Fact-checkers at the EFE wire service and Maldita traced the video within hours: it was Netanyahu's statement of 10 April 2026, his real words were about diplomacy and an "immediate price" for Israel's critics — the word "train" never appears — and the fabricated captions were attached to exploit the rawest wound in Spanish public life that week. The fakes spread anyway; debunked, they were replaced by cousins: claims that Spain had "given" Morocco €750 million for railways while its own line rotted (the real figures, checked by EFE, were refundable promotional credits for Spanish companies), claims that images from Adamuz were Urda, and a particularly ugly one claiming the minister had smiled on camera minutes after learning of the crash — a conference video timestamped four minutes after the impact, when no one outside the control room yet knew anything.

Why document the fakes at length? Because the response to them is also documented, and instructive: within forty-eight hours, the state's and the platforms' verification machinery had labelled almost every one. The disinfo wave is not a mystery to be whispered about; it is a measurable phenomenon — a real April video, recut with fake threats, riding the grief of October — and its anatomy belongs in the permanent record.

Nor was the crash season the first time individuals were marked. In January, after Adamuz, a campaign calling itself "Vindex" published the personal data of transport officials and company executives — the minister, secretaries of state, both company presidents, and notably the president of Azvi, one of

the Adamuz-renovation consortium's members — alongside the promise that "soon there will be news about Puente." The doxxing was condemned across the spectrum; its targets, chillingly, now overlap with the people this document has had to name.

The press itself, under the microscope

A final note on the information war's oldest front: accuracy. This document's research caught newspapers miscounting the Adamuz dead by twenty, caught a national outlet attributing the prior day's track failure to the wrong train number, caught transcription variants of the same recorded word — BSL, BFL, BCL — across major outlets, and caught Wikipedia's own entry momentarily contradicting itself on the death toll. None of this is scandal; all of it is the fog of the first days. It is recorded here so that the reader, meeting conflicting numbers in this very document, will understand why they exist and how they were reconciled — one figure at a time, source by source, in the appendices. And the sharpest headline of that first press cycle deserves a place in the record too: a conservative daily named the minister 'the last human error of a train driver called Pedro Sánchez' — a formulation that says everything about how the political season intended to process the crash, and nothing, this document notes, about what the evidence says.

* * *

PART TEN — THE SPEECH

We arrive now at the thread that began this document: a speech, and why it matters to a story about trains.

April 2026: "An immediate price"

On Friday, 10 April 2026, at the height of the spring's other convulsions, Benjamin Netanyahu ordered the expulsion of Spain's representatives from the Civil-Military Coordination Center at Kiryat Gat — the American-chaired body, born of the Gaza ceasefire of the previous autumn, that coordinated aid and stabilisation with the participation of some sixty countries. His reason, in his own recorded words: "Spain has smeared our heroes, soldiers of the IDF, soldiers of the most moral army in the world… I am not willing to tolerate this hypocrisy and hostility. I have no intention of allowing any country to wage a diplomatic war against us without paying an immediate price."

The trigger was Spain's stance on the war then widening across the region: its condemnation of the strikes on Lebanon as violations of international law, its foreign minister's declaration that the war on Iran was illegal, its reopening of its embassy in Tehran. And the backdrop was a diplomatic relationship that had been collapsing for two years, step by documented step: Spain's recognition of the State of Palestine in May 2025; Israel's recall of its ambassador; Spain's prohibition of arms shipments to Israel through its ports and airspace in September 2025, followed by a formal and permanent arms-sale embargo; the summoning home of Spain's own ambassador, and her formal

removal by royal decree in March 2026, leaving both embassies run by chargés. Spain's prime minister answered the expulsion the same day in Barcelona, calling for the suspension of the European Union's association agreement with Israel — "let us not allow a new Gaza in Lebanon."

That was the speech. It was, on the record, a real diplomatic rupture between two states, and it belonged to the spring of 2026.

October: the speech returns

Six months later, within hours of the Urda crash, the April video was recut, re-subtitled, and reborn as a fabricated threat of train accidents — the disinfo anatomy you have already read. And the same week, on the platform X, an automated "story" summarised the old April events as if they were breaking news, layering in claims — arms embargo, airspace closure, ambassador recall — that were, as it happens, all true, while the recirculated clip it rode on was being used elsewhere for the fake threat captions. That is the entire strange loop, documented: a real speech, about a real rupture, recycled as fake news during a real catastrophe.

Why does this document insist on including it? Because the user who commissioned this research first held up a screenshot of that recycled "story" beside the news of the crash and asked: what is going on here? The honest answer, in full, is what these two parts have laid out: the diplomatic arc was real (and is now, as of this writing, one of Europe's most frozen relationships); the recycling was disinformation; the disinformation exploited a

genuine grief; and the fact that a foreign prime minister's six-month-old speech could be made to feel relevant to Spain's deadliest rail disaster in thirteen years says something true — though this document will not say precisely what — about the informational weather in which modern catastrophes now occur.

The side-thread also has one more documented footnote worth preserving for the record: on the day of the crash itself, Israel's foreign minister, the same Gideon Sa'ar who had announced the April expulsion, sent Spain his condolences.

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PART ELEVEN — THE PATTERN: SPAIN AND ITS RAILWAYS, 1944–2026

Urda did not come from nowhere. Spanish railways carry a long memory of catastrophe, and the memory matters — because in several of the earlier cases, the questions asked then are the same questions being asked now, and in at least one documented case, a safety recommendation issued a decade ago was still, in the words of the press, "in the air."

Torre del Bierzo, 1944. In Franco's Spain, a mail train and a troop train collided in a tunnel in Palencia; the official death toll was seventy-eight, historians believe the true figure may approach five hundred, and the regime censored the news for days. The lesson of 1944 is about information: the first casualty of that disaster was the public's right to know it had happened.

El Cuervo, 1972. On 21 July 1972, on the single-track line near Seville, a four-carriage local train left its station against a red signal — forcing the exit points open as it went — and met an oncoming express at combined speed. Eighty-six people died; about forty of them were young navy conscripts on leave. The express's only fatality was a postal worker in the mail van. The railway's own investigation blamed the local train's driver, who did not survive to answer. El Cuervo held Spain's sad record for forty-one years, and its lesson — that a single human decision against a red signal can defeat an entire system of points and signals — is precisely the class of failure the LZB was later bought to eliminate. It is also, not coincidentally, the shape of the story Spain's authorities first reached for in October: one

human, one error.

Angrois, 2013. On 24 July 2013, an Alvia derailed on a curve outside Santiago de Compostela at nearly twice the speed limit; eighty of the 224 aboard died. The driver was on a phone call and passed through the emergency-brake pedal checks; the curve had no modern protection because the line's European-standard safety system ended miles earlier. For years, the case was a single word — the driver — until the investigation widened: a senior Adif safety manager, Andrés Cortabitarte, was charged and convicted with him in 2024 for approving a safety certification he himself had testified he had no competence to evaluate. Then, in January 2026 — the very week of Adamuz and Gélida — an appeals court absolved Cortabitarte and upheld the driver's conviction, raising the victims' compensation above €22 million. The European safety system whose absence was felt at Angrois was finally switched on along that stretch in August 2023: ten years after the crash. The victims' platform, which fought for the publication of a damning EU report that Madrid had delayed with a signature campaign of 267,000 names, put the philosophy that connects all these cases into one sentence at an anniversary gathering: "A train that can reach such speeds cannot depend on a human being… there are many more responsible parties in the chain, and those are the ones who must pay, politically and criminally." The driver, for his part, wrote a five-paragraph letter of apology; the victims forgave him publicly. Spain had rehearsed, in other words, every element of Urda's human-versus-system argument — and had seen it end, after thirteen years, with an appeals court.

Chamartín, 2024. The runaway train in the tunnel, saved by a deliberate human act of derailment. Its lesson is the mirror image of the others: when the system failed, improvisation saved it; the railway survived on the judgment of one employee — exactly the kind of dependence its designers thought they had engineered away, and exactly the kind of dependence that, at Urda, failed.

Place these beside the documented pattern of 2024–2026 — the derailment statistics rising from nine in 2015 to twenty-two in 2024, the autumn cluster of 2024, Adamuz, Gélida, Ricla, and five conventional lines closed or delayed across the year — and the question any reader is now entitled to ask stops being "how did this happen?" and becomes "what was the system as a whole doing?" That question is, happily, no longer rhetorical: the independent authority has a year to answer it, and this document has done its best to hand it every clue.

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PART TWELVE — AFTERMATH: MONEY, MEMORY, AND THE FUTURE

The victims' money

Within days, the money machinery that Adamuz had already built began turning for Urda. Iryo, whose compulsory insurer is AIG Europe, announced voluntary advances of €35,000 per fatality — more than the €21,000 floor the EU's passenger-rights regulation requires within fifteen days. The government's compensation package, modelled on the Adamuz decree of February 2026, was scheduled for cabinet approval on 3 November: roughly €210,000 per fatality within three months, composed of €72,000 in direct tax-free aid, €72,000 as an advance on civil liability, and a matching sum from the compulsory insurance once fault is apportioned; injury payments scaled from €2,400 to over €84,000, each likewise doubled. The legal architecture beneath is a four-lane highway of claims — the EU regulation, the mandatory traveller insurance, the state aid, and civil courts — and one of its clauses deserves special attention in this case: under the EU rules, an infrastructure manager's failure gives the train operators no force-majeure shield. Adif's error cannot excuse Renfe. Where final fault will be apportioned among them is what the lawyers are for.

The body of work

The physical work continues as this document closes. Under the judge's continuous-shift order, Actren's salvage crews work day and night; the line is expected to reopen around 14 November, seventeen days after it died — compared with the four weeks Adamuz's corridor needed in February, the new rhythm is an improvement, though more than two weeks of silence on a line that carries a region is still an eternity. Identification proceeds DNA sample by DNA sample; a state funeral is planned for mid-November in Toledo's cathedral, with the King in attendance, on the model of January's homage to Adamuz in Huelva. The airlines added capacity and capped fares; the private operators' parallel suspension ran to 2 November, the state operator's sales freeze far longer; and the drivers' strike, kept on through the mourning, ran under minimum-service decrees that still put three-quarters of high-speed trains on the road — a fact that cuts both ways. The unions demand the one change the crash made undeniable: no more "normal running" after a failure — lock the rulebook's 40 km/h (25 mph) into the machinery so that no future operator's worst day can outrun it.

The political reckoning

And the politics: a general election on 29 November, called before the crash and now reframed by it — "the railway tragedy breaks the 'love' campaign," one analysis headlined — with the main opposition party demanding an independent audit of the entire network, and the government insisting the investigation will vindicate its management. The transport minister, seven times reproached by parliament over Adamuz, once the named target of a criminal complaint, the survivor of a year in which his signature policy lurched from one closed line to the next, remains at his desk; the president of the infrastructure operator is, the press reports, on his way out; the president of the train operator, the former mayor's bus manager, is "under examination," more slowly. None of them has been charged with anything in relation to Urda, and this document states that plainly, too.

The open questions

An honest document ends with its unanswered questions, because they are the ones that will matter next. As this closes, these are the open questions the record leaves standing:

1. **Why did the computers fall?** The four-system cascade of that morning — LZB, radio, track circuits, power — has no publicly explained cause. Adif's own investigation continues; the maintenance contractor, Hitachi Rail, has been barred from the site by the court.

2. **Why did the room believe the Iryo was moving?** The recordings preserve a wrong mental picture. Whether screens, procedures, staffing, or stress produced it is the investigation's central human question.
3. **Why was the order never corrected?** The corrected-order driver proves the correction was available. Why it reached one train and not the other is the question that will decide the operator's case.
4. **Who else, if anyone?** The criminal file currently holds one silent suspect. The unions, the engineers' colleges, the victims of Adamuz and Angrois before them, and a good part of the public press all point at the system — at the missed migration, the softened deadlines, the hollowed-out oversight. A court will decide where the file stops.
5. **Will the watchdogs watch?** The new authority has three investigators and a year. The regulator is led by acting appointees after the September dismissal. The judge is twenty-nine. Whether these institutions, in their current shapes, can carry the weight now resting on them is a question this document poses but cannot answer.

A last word about the two threads

This document began with a speech and a train. They turned out to be woven together in three ways: by the fabricated video that made the speech part of the crash's aftermath; by the real diplomatic rupture the speech recorded, whose "immediate price" rhetoric is the same logic that governs blame after a disaster — someone must pay, quickly; and by the deeper symmetry that a serious reader will already have noticed: both stories are about systems that failed to replace their old machinery of control — one geopolitical, one made of cables and code — and about what rushed into the vacuum.

The speech's real words, in April, were that no country should be allowed to act against another "without paying an immediate price." The railway's real lesson, in October, is nearly the opposite: the demand for an immediate price — one operator, one error, one scapegoat — is itself the oldest failure mode in the whole catalogue. The record now belongs to the reader.

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APPENDICES

Appendix A — The day, minute by minute (all times from the public record; where sources differ, both are shown)

- **07:54** — Iryo 6076 departs Madrid Atocha (147 passengers, 5 crew).
- **08:00** — AVE 2080 departs Atocha (official count: 379 passengers, 7 crew).
- **~08:30** — LZB computers fail at Urda central; six trains halted over ~63 km (~39 miles); four-system cascade documented (LZB, GSM-R, track circuits, power).
- **08:31** — AVE driver reports partial block, code 00/000 (most restrictive).
- **08:36** — Control telephones AVE: disconnect, reconnect, "resume under BSL with normal running." Driver reads back; confirmed.
- **08:41** — Iryo driver calls: cannot restart LZB; stopped at Urda entry signal. Same "normal running" instruction given.
- **~08:43** — AVE resumes, stopped ~11 minutes at Mora.
- **08:48** — Iryo calls again: still immobilised. Room acknowledges. The 2061 (oncoming AVE) is discussed in room chatter; no warning passed to AVE 2080.
- **08:53** — Impact at PK 118+800 (official). (Contemporary reports: 08:48 Iryo timestamp; 08:53 minister; 08:54 Adif and a neighbour's photograph; 08:56 the minister's press-conference figure.)
- **08:54** — First public timestamp; emergency call to 112.

- **09:10** — Adif interrupts circulation; AITAT notified.
- **09:15** — Regional emergency coordination at maximum alert.
- **09:39** — Adif's incident report ("parte") completed, documenting the four failures.
- **10:00–16:00** — Toll climbs: 10 → 30 → 64 → 68.
- **12:15** — AITAT's three-investigator team on site.
- **~14:50** — Minister's press conference: "erroneous decision"; operator error framing; 64 dead announced jointly with regional president; 68 by 15:45.
- **21:00** — SER, El País, elDiario.es publish leaked audio simultaneously.
- **29 Oct, 10:35** — 69 confirmed (a hospital death).
- **29 Oct** — Iryo announces €35,000 advances; royals visit Toledo hospital.
- **30 Oct** — Operator interrogated as investigado; exercises right to silence.
- **31 Oct** — Judge orders round-the-clock wreckage search; first multi-victim funeral in Urda (Archbishop of Toledo presiding).
- **1 Nov** — 56 of 69 victims identified.
- **3 Nov** (scheduled) — Cabinet approves compensation package; minister appears before Congress.

Appendix B — The numbers and their disagreements (documented discrepancies)

Death toll trajectory: 10 (10:00) → 21 → 30 → 64 (13:15–14:00; splits recorded variously as 40 Iryo/24 AVE, 34/30, and one early garbled 24/6) → 68 (15:45; 44 Iryo/24 AVE, one of whom died in hospital) → 69 (29 Oct 10:35; +1 hospital death). The 68-scene + 1-hospital arithmetic was confirmed by the regional government.

Passengers: official (AITAT): AVE 379 + 7 crew; Iryo 147 + 5 crew. Other sourced figures: 378 (Renfe registers), 393 (a wire count), 397 (ministerial briefing), 297 (one newspaper, treated as an outlier). Iryo crew of five confirmed by the company.

Impact time: 8:48 (Iryo's internal timestamp), 8:53 (minister; AITAT), 8:54 (Adif, 112, a neighbour's photograph), 8:52 and 8:56 (isolated press variants). The research record treats 08:53–08:54 as canonical.

Impact speed: 160–200 km/h (100–124 mph) (AITAT estimate; exact figure pending the recorder data).

Affected stretch: 60 / 63 / 67 / 69 km (37–43 miles) in different official statements; the incident report's ticket covered km 89.5–119.7 (miles 55.6–74.4).

Trains halted by the LZB fall: six in nearly all records; one outlet said eight.

2026 rail deaths: 46 (Adamuz) + 1 (Gélida) + 69 (Urda) = 116.

Appendix C — The money, in one table

Year	Event	Document
1992	Madrid–Seville line opens with LZB; Thales lineage begins	Company records
2003–2018	Cartel operation in signalling tenders (€4.1bn)	CNMC ruling S/DC/0614/17
Jul 2019	CNMC fines: €128m total; Thales €17m; Siemens Rail €18.9m	CNMC
Dec 2019	2019 report warns old control system might not support modifications (Colpisa documents)	Heraldo/La Voz
Feb 2020	Renewal tender: €108m, 52 months, ERTMS to replace LZB	Ministry; BOE
Nov 2020	Tender collapsed over contradictory LZB-answer; desisted; Siemens had led scoring then appealed	TACRC ruling 0256/2021
Sep 2021	Award: €136.42m total (Thales €93.8m safety/ERTMS; €21.5m track circuits; UTE €21m points)	Maldita; Adif
Oct 2021	CNMC fines publicised; same weeks as award	Press
Mar 2022	EIB loan: €90m signed (project cost €640m)	EIB
May 2024	Hitachi completes €1.66bn acquisition of Thales GTS	Corporate
Jun 2024	Adif: ERTMS "advanced execution," completion "2025 and early 2026"	Adif site
Feb 2026	Contract deadline passes unmet; Adamuz follows two days later	Records
Jun 2026	ERTMS installed; in laboratory testing (per minister, post-crash)	Presser
Jun 2026	Contract extended without penalties; new horizon 2027 (Jan 2027 per minister)	IBTimes
Oct 2026	Urda	—

Programme totals: €680m (2023) → €780m (2026). EU money: €111.6m Brussels; €90m EIB; FEDER→Recovery-Mechanism switch; €5.825bn cumulative EU allocation to Adif; nine Recovery-Plan modifications 2023–2026.

Appendix D — The warnings register (consolidated)

When	Who	What
2023	European Commission	Flags line obsolescence; grants €111.6m
Aug 2025	SEMAF (letter)	Track wear; rolling-stock degradation (Adamuz stretch already 250-limited)
Jan 2026	SEMAF	Sector-wide strike call; criminal accountability demand
Spring 2026	AESF (circular)	Marcha a la vista at 40 km/h (25 mph); canton-free verification mandatory
27 Jan 2026	Vozpopuli (reporting)	Adif's own extraordinary marcha-a-la-vista orders at "controversial points"
1 Sep 2026	Minister (dismissal)	Fires transport-safety chief Báguena (no explanation given)
21 Sep 2026	Báguena (Senate, under oath)	"LZB already obsolete"; AESF never authorised Adamuz stretch
24 Sep 2026	Ouigo (Senate)	System "a bit old"; experts say replace (denies grave risk)
5 Oct 2026	Iryo CEO Favara (Senate)	LZB disconnections = his drivers' most frequent problem; reports since 2025
Sep 2026	CSIF (formal)	Excessive working hours at Adif
27 Oct ~20:30	An AVE driver (Colpisa)	Stopped 11 min at Urda; train "did not recognise the system" (train identified by El Español as AVE 2192)
Ongoing	InfoAdif public alerts	≥10 signalling incidents near Urda since 2024

Sep 2024	Opposition MPs (parliamentary register, BOCG-15-D-201)	Formal demand for a study of infrastructure vandalism and copper theft — the failure class later called "75% external"
Jan 2026	Vozpopuli (reporting from drivers and controllers)	Documents Adif's own post-Adamuz extraordinary marcha-a-la-vista orders at "controversial points" — proof the institution knew the 40 km/h (25 mph) standard

Appendix E — Dramatis personae (consolidated)

In the trains: Antonio Acedo Meneses (AVE driver, deceased); the Iryo driver (unnamed; survivor, hospitalised, declared to police). Possibly a second cab rider on the AVE (unresolved).

In the control room: the operator (interrogado, right to silence); an auxiliary; a supervisor. 16 operators on shift in the centre. The centre: ~100 posts nationally, chronic understaffing per unions.

Investigators: AITAT president Gonzalo Julián Guillén Espejo-Saavedra (naval engineer, ex–port captain); council of six with one vacancy; three field investigators + reinforcements. Judge Aristea Afroditi Ortiz Tiliakou (Orgaz No. 2, 74th promotion, in post since 3 Feb 2026); support magistrates Orgaz No. 1 and Alcázar de San Juan No. 2 (coordinator).

Companies: Adif president Pedro Luis Marco de la Peña (exit reported decided); Renfe president Álvaro Fernández Heredia (ex–Valladolid AUVASA under Mayor Puente; ex–Más Madrid councillor); Iryo CEO Fabrizio Favara; Iryo president Carlos Bertomeu. Thales España (1992 builder; 2021 contractor;

maintenance incumbent to Aug 2026); Hitachi Rail (contract inheritor since 2024; Spanish subsidiary president Pepi Andrea); Siemens (tender challenger; winner of a €1,218m train order the day before the crash).

State: Minister Óscar Puente; the dismissed safety chief Rocío Báguena; the AESF (acting leadership); former Adif presidents Isabel Pardo de Vera (UCO-implicated in the masks case) and Ángel Contreras (dismissed 2024).

Victims' voices: the Alvia 04155 platform (Angrois); the Adamuz association; families waiting at Toledo's Casa de Ejercicios with 27 psychologists.

Appendix F — The speech, fully sourced

10 April 2026: Netanyahu expels Spanish representatives from the Kiryat Gat Civil-Military Coordination Center; quotes verified via Xinhua and Spanish wire coverage; Sa'ar's "antisemitic obsession" statement; Spain's Palestine recognition (May 2025), ambassador recall (2025), arms embargo via ports/airspace (Sep 2025), permanent arms-sale embargo (Sep 2025), ambassador Salomón Pérez's formal removal (BOE, Mar 2026); Sánchez's Barcelona response (association-agreement suspension call); Israeli FM Sa'ar's condolence message on the day of Urda; EFE/Maldita debunks of the recirculated video (fake "train accident" captions; AI-modified whiteboard image).

Appendix G — A note on method and sources

This document was compiled from public reporting and official documents in Spanish and English, cross-checked where possible against primary texts (the AITAT note, the TACRC ruling, the CNMC ruling, the EU regulation, the EIB project page, parliamentary records, the ministry's blog). Authority grades applied throughout: official documents and rulings highest; wire services next; named-outlet reporting; then editorial and partisan material, always labelled as such. Items that failed verification are labelled as such in the body. The research log runs to one hundred and four separate dives; the full annotated record is archived with this document.

Nothing here is legal advice, an accusation, or a verdict. It is a map of everything known, offered so that anyone — friend, journalist, judge, or simply a citizen who rides the train — can find their way through the worst railway disaster of Spain's century so far.

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Written in one continuous pass from the public record, 1–2 November 2026.